

# Network Theory (BAECE103)

## Course Introduction

21/07/2026

Dr. Sudipta Kumar Sarkar

*Assistant Professor (Research), SENSE*



School of Electronics Engineering (SENSE)  
VIT, Vellore, India

# Faculty Introduction



Course Code: BAECE103

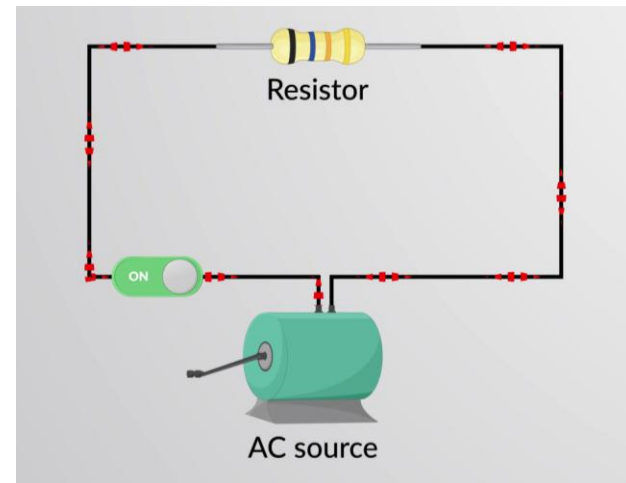
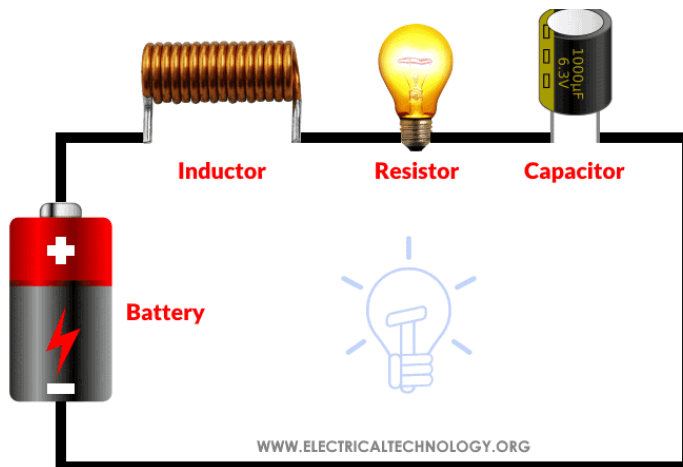
L T P C : 3 1 0 4

Faculty: Dr. Sudipta Kumar Sarkar  
Assistant Professor (Research)  
School of Electronics Engineering (SENSE)

Faculty Room: CBMR 201G

Mobile: 936186759

Open hours: Monday : 15:00 to 16:00 hrs.  
Wednesday: 15:00 to 16:00 hrs.



# Course Objectives



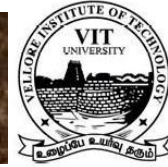
- 1.To familiarize circuit laws and theorems and to solve DC and AC circuits.
- 2.To understand mathematical methods such as the Laplace transform, Fourier transform and graph theory for solving circuit problems.
- 3.To develop an ability to analyze two-port networks, first and second-order circuits

# Course Outcomes



1. Compute DC circuit parameters using circuit laws and theorems
2. Determine the natural and step responses of RC, RL, and RLC circuits
3. Apply appropriate techniques and theorems for sinusoidal steady-state analysis
4. Apply the Laplace transform to circuits for steady-state and transient analysis
5. Compute two-port network parameters and network graph matrices for circuit analysis

# Syllabus



|                                                                                                                                                                                                                                                                                                                                                                                          |                                                              |   |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---|-------|
| Module:1                                                                                                                                                                                                                                                                                                                                                                                 | <b>DC Circuit Analysis and Theorems</b>                      | 9 | hours |
| Basic Circuit Elements and Sources; Ohm's Law; Kirchhoff's Laws; Series Resistors and Voltage Division; Parallel Resistors and Current Division; Wye-Delta Transformations; Methods of Analysis-Nodal Analysis, Mesh Analysis; Theorems-Superposition Theorem, Source Transformation Theorem, Thevenin's Theorem, Norton's Theorem, Maximum Power Transfer Theorem, Reciprocity Theorem. |                                                              |   |       |
| Module:2                                                                                                                                                                                                                                                                                                                                                                                 | <b>Transient Analysis of First and Second-Order Circuits</b> | 9 | hours |
| First-Order Circuits- Source-Free RC Circuit, Source-Free RL Circuit, Singularity Functions, Step Response of an RC Circuit, Step Response of an RL Circuits; Second-Order Circuits- Source-Free Series RLC Circuit, Source-Free Parallel RLC Circuit, Step Response of a Series RLC Circuit, Step Response of a Parallel RLC Circuit, General Second-Order Circuits.                    |                                                              |   |       |

# Syllabus



|                                                                                                                                                                                                                                                                                                                                                                                    |                                         |   |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---|-------|
| Module:3                                                                                                                                                                                                                                                                                                                                                                           | <b>Sinusoidal Steady State Analysis</b> | 9 | hours |
| Sinusoids; Phasors; Phasor Relationships for Circuit Elements; Impedance and Admittance; Kirchhoff's Laws in the Frequency Domain; Instantaneous and Average Power; Complex Power; Methods of Analysis-Nodal Analysis, Mesh Analysis; Theorems-Superposition Theorem, Source Transformation Theorem, Thevenin's Theorem, Norton's Theorem, Maximum Average Power Transfer Theorem. |                                         |   |       |
| Module:4                                                                                                                                                                                                                                                                                                                                                                           | <b>Circuit Analysis in the S-domain</b> | 9 | hours |
| Circuit Element Models; Circuit Analysis with and without initial conditions; Transfer Functions; From Laplace to Fourier Transform- Frequency Domain Analysis of Linear Circuits; Passive Filters- Lowpass Filter, High Pass Filter, Bandpass Filter, Band Stop Filter.                                                                                                           |                                         |   |       |

# Syllabus



|                                                                                                                                                                                                                                                            |                                             |   |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---|-------|
| Module:5                                                                                                                                                                                                                                                   | <b>Two-Port Networks and Network Graphs</b> | 7 | hours |
| Two-Port Networks-Impedance Parameters, Admittance Parameters, Hybrid Parameters, Transmission Parameters, Relationships Between Parameters, and Interconnection of Networks. Network Graphs-Incidence, Tie-Set, and Cut-Set Matrices for Circuit Analysis |                                             |   |       |

|           |                            |   |       |
|-----------|----------------------------|---|-------|
| Module: 6 | <b>Contemporary Issues</b> | 2 | hours |
|           |                            |   |       |

**Total Lecture Hours: 45 hrs**

**Tutorial Hours: 15 hrs**

# Books



## Text Books:

- |    |                                                                                                                                                |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Charles K. Alexander, Matthew N. O. Sadiku, “Fundamentals of Electric Circuits,” 5th Edition, Tata McGraw Hill Education Private Limited, 2012 |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|

## Reference Books:

- |    |                                                                                                              |
|----|--------------------------------------------------------------------------------------------------------------|
| 1. | W.H. Hayt, J.E. Kemmerly and S.M.Durbin, “Engineering Circuit Analysis”, 7th Edition, TMH, New Delhi, 2010   |
| 2. | Allan R. Hambley, “Electrical Engineering – Principles & applications”, Pearson Education, 4th Edition, 2008 |

# Mode of Evaluation:



| Mode           | Max. Mark | Weightage (%) |
|----------------|-----------|---------------|
| CAT I          | 50        | 15            |
| Quiz 1         | 10        | 10            |
| CAT II         | 50        | 15            |
| Quiz 2         | 10        | 10            |
| Course Project | 10        | 10            |
| FAT            | 100       | 40            |
| <b>Total</b>   |           | <b>100</b>    |

# Lecture 1/Module 1

Introduction, DC Circuit Analysis and Theorems

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# Lecture Topics



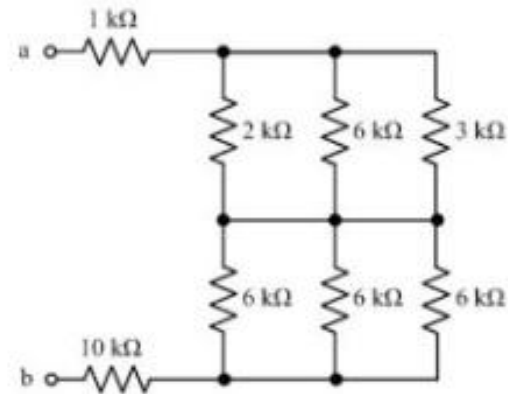
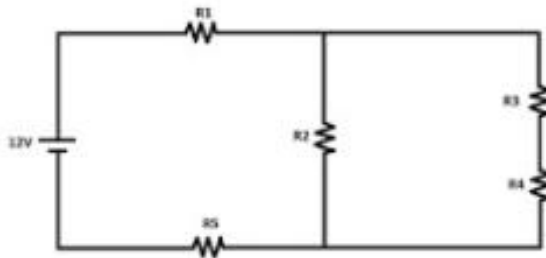
- Basic Circuit Elements and Sources;
- Ohm's Law; Kirchhoff's Laws
- Series Resistors and Voltage Division;

# Network and Circuit



## DIFFERENCE BETWEEN

## CIRCUIT **V/S** NETWORK



**Note:** Electric networks haven't any closed path whereas electric circuits have closed path

# Network Theory



Theory → Analysis + Synthesis

In this course we will cover only Analysis

# Circuit Elements



Resistor



Capacitor



Inductor



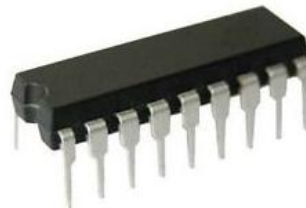
Diode



LED



Transistor



IC



Relay

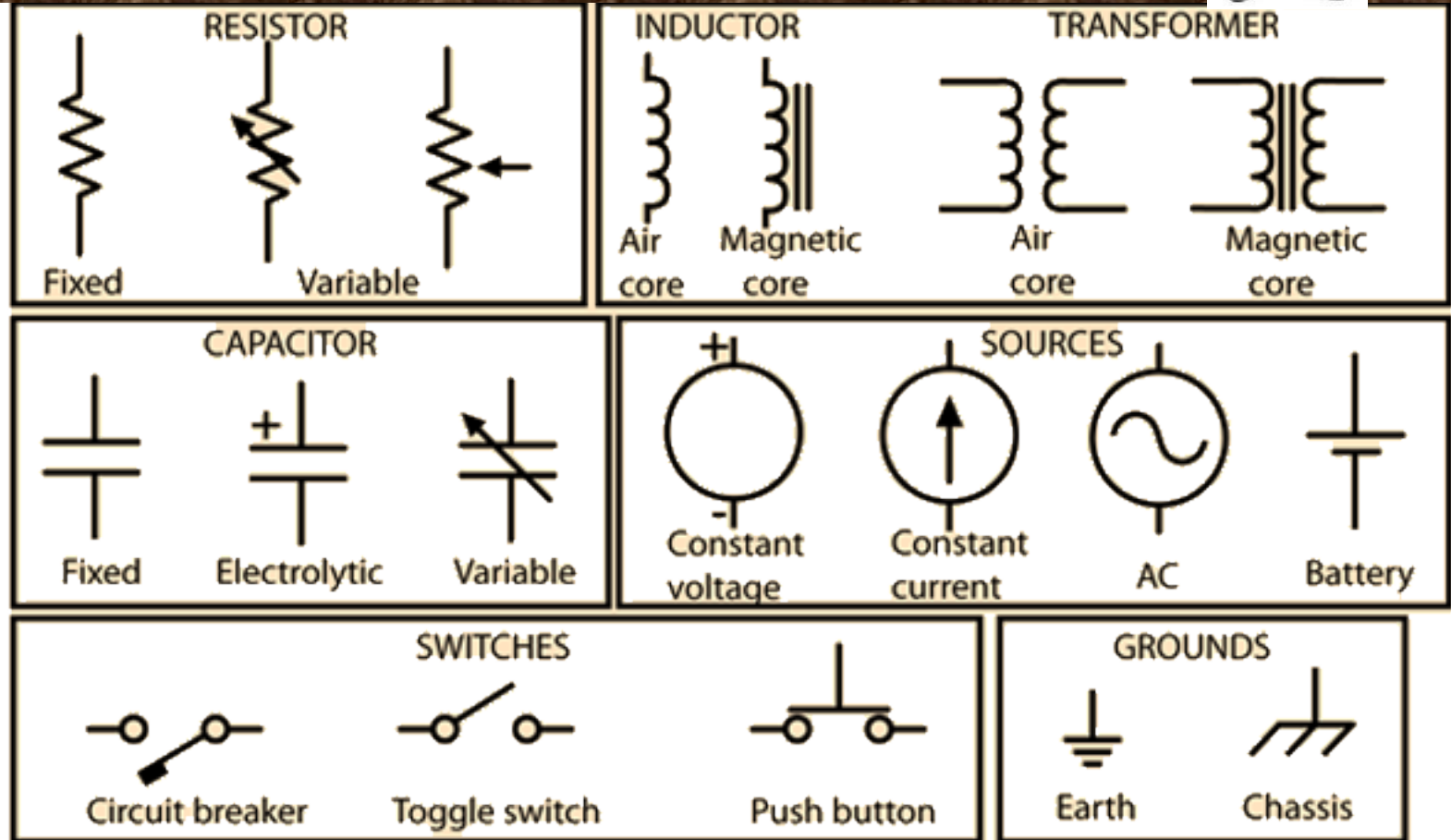


Battery

Basic Electrical Components used for Circuit Designing

[www.TheEngineeringProjects.com](http://www.TheEngineeringProjects.com)

# Circuit Symbols



Thank You